

22 Analytical techniques

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Infrared absorption helps identify functional groups; a broad O-H band differs for alcohol and carboxylic acid. Mass spectra provide molecular-ion mass and fragmentation clues; isotope patterns can indicate halogens. ^1H NMR signals reflect chemically distinct proton environments, with integration showing relative counts and splitting from neighbouring protons.

Worked example: follow the reasoning

An M and M+2 peak of near 3:1 intensity suggests one chlorine atom because ^{35}Cl and ^{37}Cl are abundant isotopes.

Quick check · multiple choice

A Which IR feature suggests an O-H group? [1]

- A Only a ^{13}C NMR peak
- B Broad O-H absorption
- C A chlorine isotope pair
- D Absence of all absorption

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A compound shows molecular ion m/z 46 and broad O-H IR absorption; ^1H NMR has signals integrating 3:2:1. Suggest a structure and explain integration. [3]

Answers and explanation

Quick check: B. Broad O-H absorption

Question 1 [3 marks]

Ethanol $\text{CH}_3\text{CH}_2\text{OH}$ [1]; CH_3 gives 3 H, CH_2 gives 2 H, OH gives 1 H [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.